

ShopFlow AWS Project

S3 Object Storage Design

Buckets - Versioning - Lifecycle - Encryption - CORS

Module 3 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon S3 is the primary object storage layer for ShopFlow. It stores application artifacts, CI/CD pipeline outputs, RDS exports, product images, and the compiled React SPA. S3 has no VPC dependency and is deployed before any compute resource.

- Globally unique bucket names -- environment suffix prevents collisions
- All buckets private by default -- product images served via CloudFront only
- AES-256 server-side encryption on every bucket at no extra cost
- Versioning enabled on artifacts and CI/CD buckets for rollback capability
- Lifecycle rules move objects to cheaper storage classes automatically

2. Buckets

Bucket Name	Purpose	Versioning	Lifecycle
shopflow-artifacts-dev	Spring Boot JARs, build outputs	Enabled	Delete old versions after 30 days
shopflow-cicd-artifacts-dev	CodePipeline stage artifacts	Enabled	Expire after 7 days
shopflow-rds-exports-dev	Aurora snapshot exports, SQL dumps	Enabled	IA 30d, Glacier 90d, Delete 365d
shopflow-product-images-dev	Product photos (served via CloudFront)	Disabled	Noncurrent delete after 90 days
shopflow-spa-dev	React + Vite frontend build output (dist/)	Disabled	None

3. Storage Class Lifecycle (RDS Exports Bucket)

Lifecycle transitions automatically move objects to cheaper storage:

Day	Storage Class	Cost / GB / Month	Retrieval Speed
0	S3 Standard	\$0.023	Immediate
30	S3 Standard-IA	\$0.0125	Same speed + \$0.01/GB fee
90	S3 Glacier Instant	\$0.004	Milliseconds + \$0.03/GB fee
365	DELETED (auto-expiry)	--	--

4. Security Configuration

4.1 Public Access Block

All four Public Access Block flags are enabled on every bucket. This prevents any object from being publicly accessible, even if an IAM policy or ACL accidentally grants public access.

4.2 Encryption

SSE-S3 (AES-256) is applied as the default encryption configuration on all buckets. Encryption is automatic and free -- objects are encrypted before being stored.

4.3 CORS (Product Images Bucket)

CORS Setting	Value
Allowed Origins	http://localhost:5173 (dev), https://shopflow.example.com (prod)
Allowed Methods	GET, PUT, POST
Allowed Headers	* (all headers)
Max Age	3600 seconds

5. CloudFormation Outputs

Export Name	Value
shopflow-artifacts-bucket	shopflow-artifacts-dev
shopflow-cicd-bucket	shopflow-cicd-artifacts-dev
shopflow-rds-exports-bucket	shopflow-rds-exports-dev
shopflow-images-bucket	shopflow-product-images-dev
shopflow-spa-bucket	shopflow-spa-dev